

COM3502-4502-6502 SPEECH PROCESSING

Lecture 12

The Nature of Speech



COM3502-4502-6502 Speech Processing: Lecture 12, slide 1

1

Speech is not just Audible Text



Speech is ...

- continuous
- variable
- ambiguous
- effortful
- contrastive
- prosodic
- adaptive
- context-dependent
- meaningful
- referential
- indexical
- rhetorical
- paralinguistic
- personalised
- affective
- multimodal
- contaminated

communicative

embodied

ostensive

pragmatic

situated

grounded

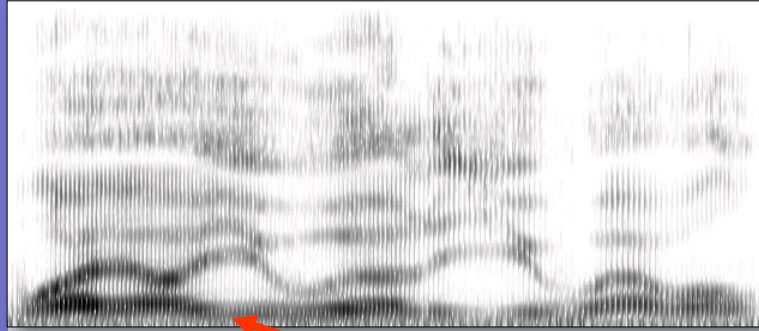


COM3502-4502-6502 Speech Processing: Lecture 12, slide 2

2

Speech is ... *continuous*

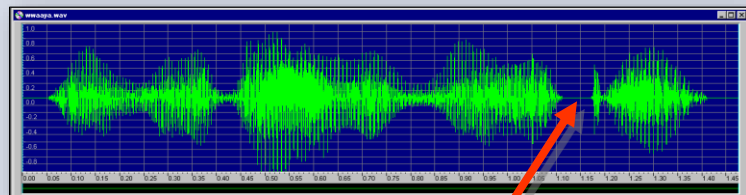
“Why are you early you owl?”



Can't see where one word stops and another begins (*there are no gaps*)

Speech is ... *continuous*

“We were away a year ago.”



The only silence is the 'g' in "ago" !

Speech is ... *variable*

العربي български català 中国话 hrvatski český
english ελληνικά עברית हिंदी italiano 日本語
한국어 românește русский српски ภาษาไทย



There are >7,800,000,000 people
in the world

... and they all speak differently

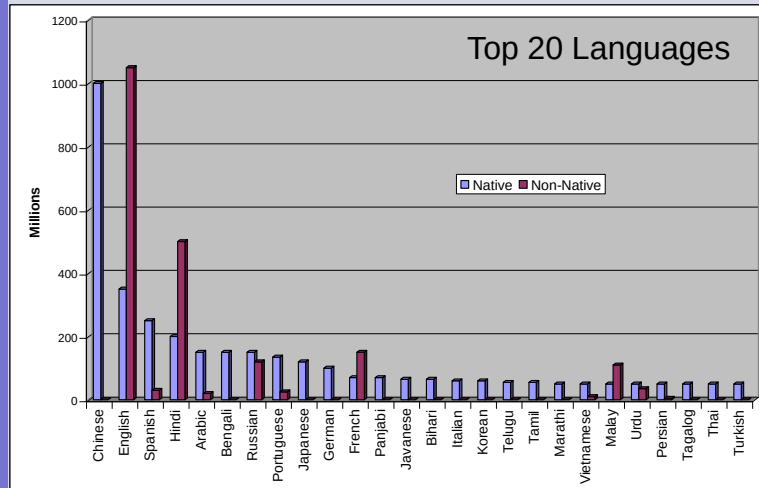
Speech is ... *variable*



There are 3000-8000 languages:

- 6500 'living' languages
- 83% are limited to single countries
- 52% are spoken by less than 10,000 people
- 28% are spoken by less than 1000 people
- According to the National Geographic Society, one language dies out every two weeks
- Half of the world's languages will be out of use by 2100

Speech is ... *variable*



Speech is ... *variable*



- Within any given language, groups of individuals use local dialects and accents
- A '**dialect**' is when different words are used
 - “*snotter clot*” is Geordie (*Newcastle*) for “*handkerchief*”
 - “*china*” is Cockney rhyming slang for “*mate*” (short for “*china plate*”)
 - “*beck*” is Yorkshire (*Old Norse*) for “*stream*”
- An '**accent**' is when different sounds are used
 - “*bath*” and “*path*” (rhymes with “*cat*” in the north of England)
 - “*bath*” and “*path*” (rhymes with “*car*” in the south of England)
- Accents and dialects reflect regional and/or social differences

Speech is ... *variable*

The British Isles

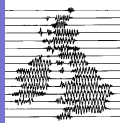


Regional accents of:

"When a sailor in a small craft faces the might of the vast Atlantic Ocean today, he takes the same risks that generations took before him"



ABI



Accents of the British Isles



COM3502-4502-6502 Speech Processing: Lecture 12, slide 9

9

Speech is ... *variable*



'Between'

- Even within a given language/dialect/accent group, different individuals sound different

- 'Inter-speaker variation' is caused by:

- age (e.g. child, teenager, adult, senior citizen)
- gender
- physical characteristics (e.g. vocal tract size)
- social habits

'Within'

- 'Intra-speaker variation' can arise from:

- physiological factors (e.g. bad health, fatigue)
- psychological factors (e.g. mood, emotion)
- external factors (e.g. the environment)



COM3502-4502-6502 Speech Processing: Lecture 12, slide 11

11

Speech is ... *variable*



“fish and chips” → “fish ’n’ chips”

“temporary” → “tem’pry”

“can be” → “cam be”

“bread and butter” → “bre’m butter”

This is not ‘sloppy’ speaking!

This is a normal part of communicative behaviour that balances the demands of *speaking* against the demands of *listening*

Speech is ... *variable*



- People can (*choose to*) speak:

- loudly *or* softly
- quickly *or* slowly
- clearly *or* mumbled
- formally *or* casually

... and this can change in the course of a single utterance

- There is also inherent variation
 - even a trained phonetician cannot produce the same word in *exactly* the same way on different occasions

Speech is ... *variable*



Question ...

What 'external factors' might influence the way that a person speaks?

Speech is ... *variable*

E
x
t
e
r
n
a
l

F
a
c
t
o
r
s



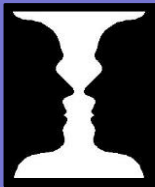
- Noise
 - hyper-articulation
 - the 'Lombard Effect'
- Vibration
 - chest
 - vocal tract
- The task
 - casual conversation
 - reading out loud
 - lecturing
- The listener
 - a child ('parentese')
 - a non-native person
 - a hearing-impaired individual
 - an animal
 - a machine (!)
- The cognitive load
 - interaction with other tasks
 - stressful/emotional situations
- Alcohol/drugs
 - impaired motor control

Speech is ... *ambiguous*



- Variability means that signals we'd like to be the same are actually quite different ...

speech speech **speech** *speech*
speech



- Ambiguity arises when signals we'd like to be different end up being the same ...

“to” vs. *“two”* vs. *“too”*
“hear” vs. *“here”*

← 'homophones'

Speech is ... *ambiguous*

“four candles” vs. *“fork handles”*

“great ape” vs. *“grey tape”*

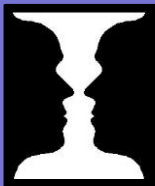
“law and order” vs. *“Laura Norder”*

“Joe is office head” vs. *“Joe is off his head”*

“This nudist play will wreck a nice beach”

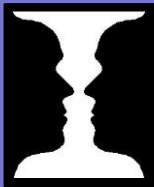
vs.

“This new display will recognise speech”



Speech is ... *ambiguous*

- Do you always understand lyrics ???



@bechillcomedian

<https://youtu.be/nEdR44Ifb4>

23

Speech is ... *language+*

What's
being
said

How it's
being
said

Other
speaker-
generated
behaviour

- Rich in '*linguistic*' information
 - “*that isn't it, is it?*”
 - “*what's in the road ahead?*” vs. “*what's in the road, a head?*”
- Rich in '*para-linguistic*' information
 - individuality
 - attitude
 - emotion
- Rich in '*extra-linguistic*' information
 - breathing noises
 - lip-smacks

24

Speech is ... *d'd'dis-disfluent*

- Normal spontaneous speech contains:
 - false-starts
 - repeats
 - filled pauses (“*uhms*” and “*errs*”)
 - non-linguistic utterances (NLUs)
 - overlaps
- Spoken utterances are not as ‘well-formed’ as written sentences
- So-called ‘disfluencies’ actually make speech easier (*for a human*) to produce and understand

Speech is ... *contaminated*

- Everyday speech occurs in normal environments (*not recording studios*) which include:
 - other voices
 - competing sounds
 - noise & distortion
 - reverberation



- It's a bit like this ...

“xn xpplx x dxy kxxps thx dxctxr xwxy”

... but, it's much more like this:

“ax axxe a xax xeexx xxe xxxex axax”

This slide set has covered ...

- Why speech is an *interesting* signal to process
- Why speech is a *challenging* signal to process ...
 - continuity
 - variability & ambiguity
 - linguistic richness
 - emotion & disfluency
 - multimodality & contamination
 - multi-disciplinarity

Any Questions ?



Next time ...

Sounds and Symbols